PROJECT REPORT

Library Crowd Monitoring &

Smart Lighting System

ESP32 · IR Sensors · LCD I2C · Relay Module

1. Project Overview

The Library Crowd Indicator System is designed to monitor the number of people inside a library and display the current occupancy status in real time. The system uses two infrared (IR) sensors — one at the entrance and one at the exit — to count people entering and leaving.

An ESP32 microcontroller processes the sensor signals, maintains a live occupancy count, and displays both the count and the crowd status on a 16×2 LCD screen via I2C. This allows users and library administrators to instantly assess whether the library is empty, moderately occupied, or crowded.

2. Objectives

- Monitor the number of people inside the library in real time
- Provide a clear crowd level indication: Empty, Medium, or Crowded
- Help users know the occupancy status before entering the library
- Assist library staff in managing and monitoring visitor flow
- Deliver a low-cost, embedded-systems solution with minimal infrastructure
- Improve organization and reduce overcrowding in study spaces

3. New Feature — Automated Smart Lighting

> **Smart Lighting Control is a new feature added on top of the core crowd monitoring system.**

Overview

Instead of leaving library lights running at all times — regardless of whether anyone is inside — the Smart Lighting feature ties lighting control directly to the real-time occupancy count. The relay module acts as the bridge between the ESP32's digital logic and the mains lighting circuit.

Why We Added This Feature

- Energy savings: Lights only consume electricity when the library is actually occupied.
- Fully automatic: No switches, no manual intervention — lighting responds instantly to the count.
- Environmental impact: Reduced unnecessary electricity consumption improves sustainability.
- User experience: Staff and visitors always walk into a lit library — no fumbling for switches.

How It Works in the Code

After every count update, the loop checks whether count > 0. If true, GPIO 26 goes LOW and the relay closes the lighting circuit. If count reaches 0, GPIO 26 goes HIGH and the relay opens, cutting power to the lights. The logic is intentionally simple and reliable:

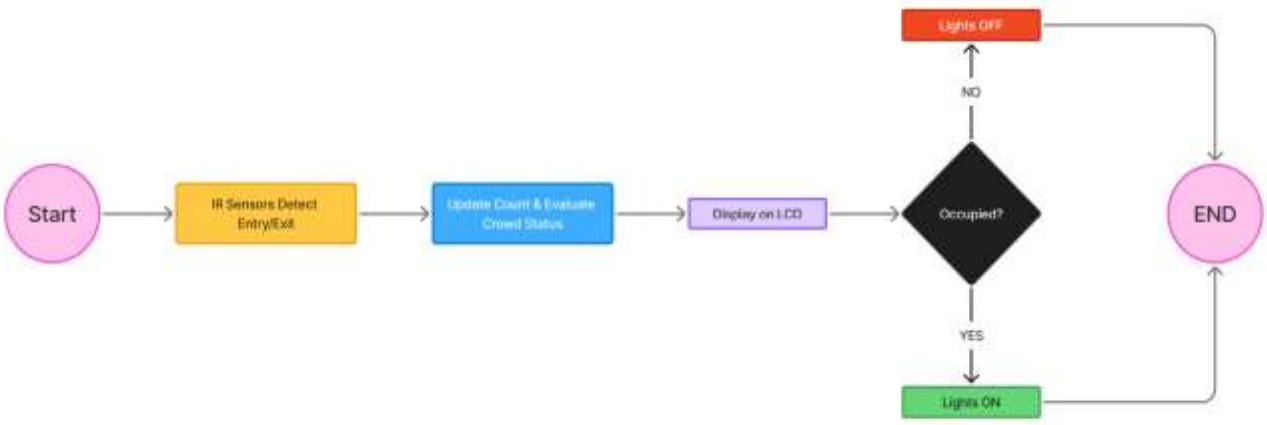
```
if (count > 0) {
    digitalWrite(RELAY_PIN, LOW);   // Lights ON
} else {
    digitalWrite(RELAY_PIN, HIGH);  // Lights OFF — library empty
}
```

4. System Architecture

The system consists of three main modules:

- Input: Two IR sensors (Entry and Exit).
- Processing: ESP32.
- Output: LCD Display, Relay Module (turns lights ON and OFF).

5. System Flowchart



6. System Components



ESP32
Microcontroller Board



IR Sensor
Object Detection



LCD I2C
Display Module



Relay Module
Power Switching



LithiumWerks Battery
Power Source

Parameter	Details
ESP32	Main microcontroller — processes all sensor data and drives outputs
IR Sensor (Entry)	Detects people entering the library (GPIO 27)
IR Sensor (Exit)	Detects people leaving the library (GPIO 25)
LCD 16×2 (I2C)	Displays live count and crowd status (SDA: GPIO 33, SCL: GPIO 32)
Relay Module	Controls the library lighting based on occupancy count (GPIO 26)
Lithium Battery	Rechargeable power source for the system

6.1 ESP32 — Microcontroller Board

The ESP32 is the brain of the system. It reads signals from both IR sensors, maintains the occupancy counter, updates the LCD display, and controls the relay for automated lighting. Operating at 3.3V logic with dual-core processing, it handles all real-time decision-making efficiently.

6.2 IR Sensors — Entry & Exit Detection

The system uses two active IR sensors. Each sensor emits an infrared beam and detects when the beam is interrupted by a passing person. This makes them ideal for precise people-counting at a single doorway.

Parameter	Details
Detection Type	Active IR — detects beam interruption (not heat)
Output Signal	Digital LOW when beam is interrupted, HIGH when clear
Entry Sensor Pin	GPIO 27
Exit Sensor Pin	GPIO 25
Supply Voltage	3.3V / 5V
Detection Range	A few centimetres to several metres (model-dependent)
Field of View	Narrow — 35°–60°, ideal for doorway beam alignment
Mounting Height	0.5 – 1.5 metres depending on application

6.3 LCD Display (16×2) with I2C Module

The 16×2 LCD screen displays two lines of information: the current people count and the occupancy status (Empty / Medium / Crowded). The I2C backpack module (PCF8574) reduces wiring to just two GPIO pins — SDA and SCL — making integration clean and straightforward.

Parameter	Details
Display Size	16 columns × 2 rows
Interface	I2C (2-wire: SDA + SCL)
I2C Address	0x27 (default)
SDA Pin	GPIO 33
SCL Pin	GPIO 32
Backlight	Enabled — always on

6.4 Relay Module — Smart Lighting Control

The relay acts as an electrically controlled switch between the ESP32 (3.3V logic) and the mains-powered lighting system. When the library has at least one person present, the ESP32 sends a LOW signal to the relay, closing the circuit and turning the lights on. When the count reaches zero, the relay opens and lights turn off.

Parameter	Details
Control Pin	GPIO 26
Trigger Logic	LOW signal = relay activates (lights ON)
Default State	HIGH (relay off — lights OFF when library is empty)
Control Terminals	VCC, GND, IN
Load Terminals	COM, NO (Normally Open), NC (Normally Closed)

6.5 Lithium Battery — Power Supply

A rechargeable lithium battery is used to power the system instead of a traditional 9V battery. Lithium batteries provide a more stable voltage and higher current, which is important for reliable ESP32 operation. They also last longer and are rechargeable, making them more efficient and suitable for embedded and IOT application.

7. How the System Works

The system operates through the following continuous loop:

- Both IR sensors monitor the library entrance and exit at all times.
- When a person crosses the entry sensor (GPIO 27 goes LOW), the people counter increments by 1.
- When a person crosses the exit sensor (GPIO 25 goes LOW), the counter decrements by 1 (minimum 0).
- After each change, the LCD updates instantly to show the new count and crowd status.
- A 500ms debounce delay prevents double-counting from a single crossing event.
- If count > 0, the relay (GPIO 26) receives LOW and the lights turn ON automatically.
- If count = 0, the relay receives HIGH and the lights turn OFF — saving energy.

8. Occupancy Status & Lighting Behaviour

The system classifies library occupancy into three levels, each displayed on the LCD and mapped to a lighting action:

Count	Status	Lighting Behaviour
0	**Empty**	Lights OFF — no occupants detected
1 – 3	**Empty**	Lights ON — minimal occupancy
4 – 7	**Medium**	Lights ON — moderate occupancy
8+	**Crowded**	Lights ON — library is full

9. Full Source Code

The complete firmware is written in Arduino C++ for the ESP32 platform using the LiquidCrystal_I2C library for display control.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// ── Pin Definitions ──────────────────────────
const int IR_ENTRY   = 27;   // Entry IR sensor
const int IR_EXIT    = 25;   // Exit IR sensor
const int RELAY_PIN  = 26;   // Relay (lighting control)

const int SDA_PIN    = 33;   // LCD I2C data
const int SCL_PIN    = 32;   // LCD I2C clock
```

```
// ── State Variables ──────────────────────────────
int count = 0;
LiquidCrystal_I2C lcd(0x27, 16, 2);

// ── Setup ────────────────────────────────────────
void setup() {
  Serial.begin(115200);
  Wire.begin(SDA_PIN, SCL_PIN);

  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Library System");
  delay(2000);

  updateDisplay();

  pinMode(IR_ENTRY,  INPUT);
  pinMode(IR_EXIT,   INPUT);
  pinMode(RELAY_PIN, OUTPUT);

  digitalWrite(RELAY_PIN, HIGH); // Start with lights OFF
}

// ── Main Loop ────────────────────────────────────
void loop() {
  bool entryDetected = digitalRead(IR_ENTRY) == LOW;
  bool exitDetected  = digitalRead(IR_EXIT)  == LOW;

  // Person enters
  if (entryDetected) {
    count++;
    updateDisplay();
    delay(500);                          // Debounce
    while(digitalRead(IR_ENTRY) == LOW); // Wait for clear
  }

  // Person exits
  if (exitDetected) {
    if (count > 0) count--;
    updateDisplay();
    delay(500);
    while(digitalRead(IR_EXIT) == LOW);
  }

  // Smart Lighting: ON when occupied, OFF when empty
  if (count > 0) {
    digitalWrite(RELAY_PIN, LOW);   // Lights ON
  } else {
    digitalWrite(RELAY_PIN, HIGH);  // Lights OFF
  }
```

```
  delay(100);
}

// ── Display Update ──────────────────────────
void updateDisplay() {
  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("Count: ");
  lcd.print(count);

  lcd.setCursor(0, 1);
  if      (count >= 0 && count <= 3) lcd.print("Status: Empty");
  else if (count >= 4 && count <= 7) lcd.print("Status: Medium");
  else if (count >= 8)               lcd.print("Status: Crowded");
}
```

Code Walkthrough

setup() — Runs once at startup:

- Initialises the I2C bus on custom pins (SDA: 33, SCL: 32) using Wire.begin().
- Initialises the LCD, turns on backlight, and shows a welcome message for 2 seconds.
- Sets IR sensor pins as INPUT and relay pin as OUTPUT.
- Sets relay HIGH initially — lights off, library assumed empty.

loop() — Repeats every 100ms:

- Reads both IR sensors. A LOW reading means the beam is broken (person detected).
- Entry detection increments count; exit detection decrements it (floor at 0).
- After each event, a 500ms delay prevents bouncing, then waits for the beam to clear.
- Lighting is updated after every loop cycle based on the current count value.

updateDisplay() — Called after every count change:

- Clears the LCD and reprints the current count on line 1.
- Determines and prints the crowd status label (Empty / Medium / Crowded) on line 2.

10. Pin Mapping Summary

Variable	ESP32 Pin	Function
`IR_ENTRY`	GPIO 27	Entry IR sensor output
`IR_EXIT`	GPIO 25	Exit IR sensor output
`RELAY_PIN`	GPIO 26	Relay control (lighting)
`SDA_PIN`	GPIO 33	LCD I2C data line
`SCL_PIN`	GPIO 32	LCD I2C clock line

11. Future Improvements

The system can be enhanced in several ways to improve functionality and usability:

- Add a card-based entry system to track user stay duration and estimate when seats will likely become available.
- Add a buzzer to alert occupants before lights turn off.
- Expand coverage to include multiple entrances or corridors in the library.
- Integrate with a mobile application to monitor library occupancy remotely.
- Connect to Wi-Fi or IoT platforms for real-time online updates.
- Improve sensor accuracy or use multiple sensors to prevent counting errors.

12. Conclusion

The Library Crowd Monitoring and Smart Lighting System provides a practical, real-time solution for tracking and displaying library occupancy. By combining accurate IR-based people counting with an intuitive LCD interface, the system gives both visitors and staff immediate visibility into the library's crowd level.

The addition of automated smart lighting elevates the project beyond basic monitoring — turning it into an energy-aware, fully hands-free environment. Lights respond instantly to actual occupancy, eliminating waste and improving the experience for everyone who uses the space.